

On-boarding and retaining maintainer talent for MNE-Python

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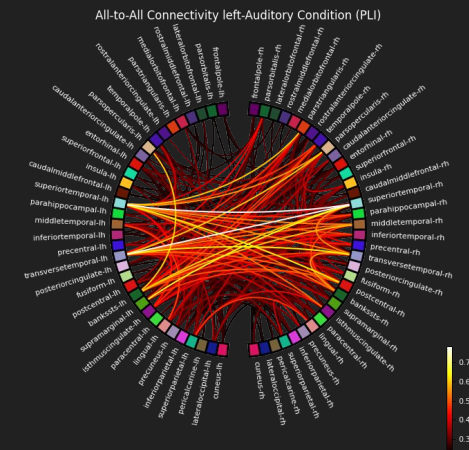
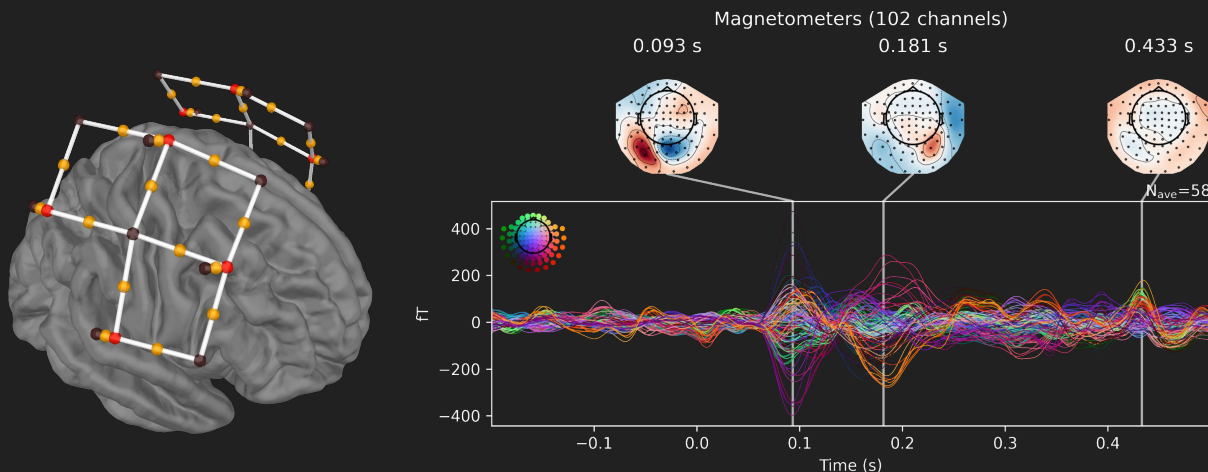
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What is MNE-Python?

- I/O, data cleaning, analysis & visualization for neuroimaging: MEG, EEG, ECoG, fNIRS...
 - ~15 org-managed extension packages
 - ~50 community-contributed extension packages



Maintainers are hard to find

Problem: Academia's incentive structure

Work on open-source scientific software not valued highly

- Doesn't improve with seniority
- Drives contributor/maintainer attrition
- Reinforces biases found in broader open-source community

Westner *et al* 2025. **Cycling on the Freeway: The perilous state of open-source neuroscience software.** *Imaging Neuroscience*. doi:10.1162/imag_a_00554

Problem: Domain knowledge required

- Can take years to accrue
- Hard to impart piecemeal
- Software maintenance skills must be learned too

If that wasn't bad enough...

- Contributing to MNE-Python was getting harder
- Founder's lab closing
- Existing maintainers leaving

Addressing the crisis

- Contributing getting harder → New Developer sprints
- Founder's lab closing → Intermediate sprints
- Existing maintainers leaving → Intensive onboarding

New Developer Sprints

- 2021 March, 2022 July
 - 1 week (30 hours), online (flexible time-of-day)
 - **PAID** (thank you CZI EOSS)
- Application guaranteed a working dev environment
 - change directory to the root of your repository clone, and paste the output of the command ``git remote -v``.
 - start an IPython session and run the commands ``import mne; mne.sys_info()`` and paste the output here.

New Developer Sprints

- Pre-selected issues (several per participant)
 - Reinforce the contribution cycle
(fetch → branch → write → test → commit → push → PR → CIs → reviews)
 - Learn rST, linting, tooling, etc
Without the cognitive overhead of writing neuroscience code
- Short “career” talks from current/former contributors/maintainers

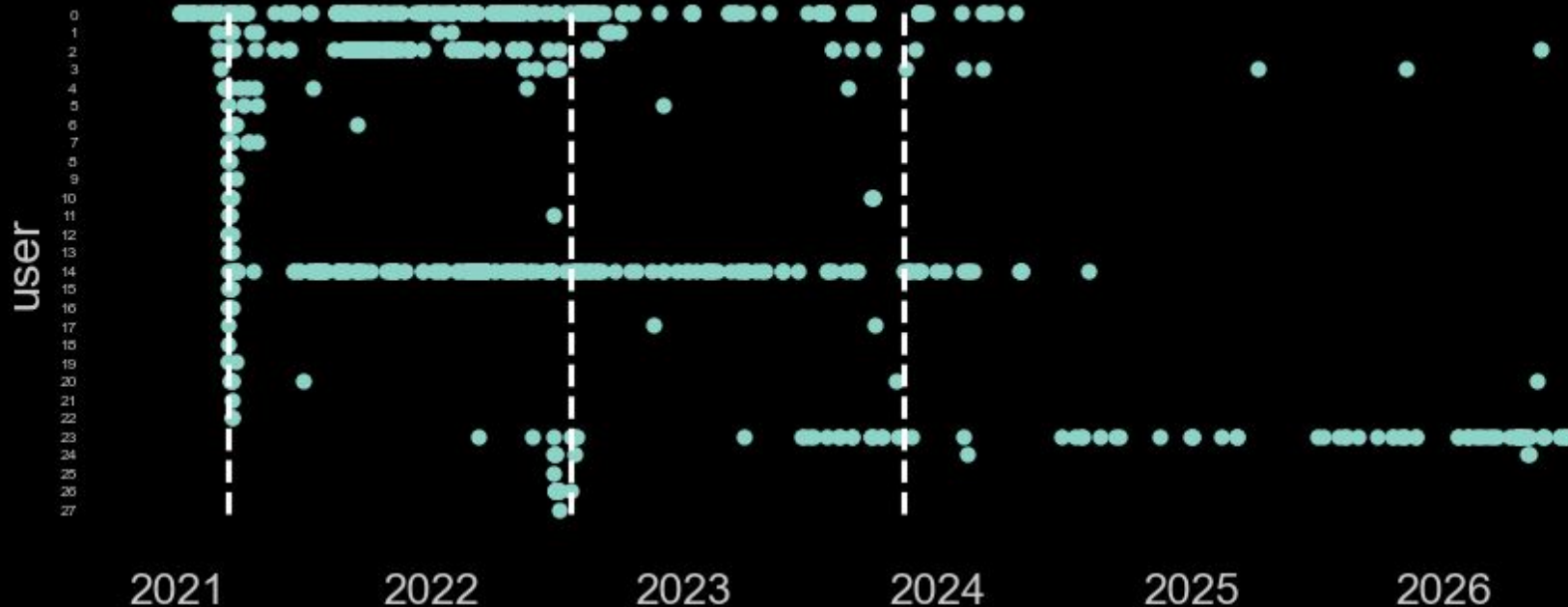
Intermediate Sprint

- 2023 November
 - 1 week (30 hours), online (flexible time-of-day)
 - **PAID** (thank you CZI EOSS)
- Open to alumni of New Dev Sprints
 - Pre-selected larger projects
 - Daily short tutorials on CIs, debugging, etc

Maintainer Onboarding

- 2025 June - 2027 May
 - 2 years (10 hours/week)
 - **PAID** (thank you NSF POSE)
- First 6 months: best-practice “lessons” & homework
 - Optimization, code review, CIs, testing, environments, dependencies, packaging, releases
- Second 6 months: PR triage/review, pair programming
 - Big-ish doc or test refactoring, CI upgrades / automation

Preliminary Results: Retention



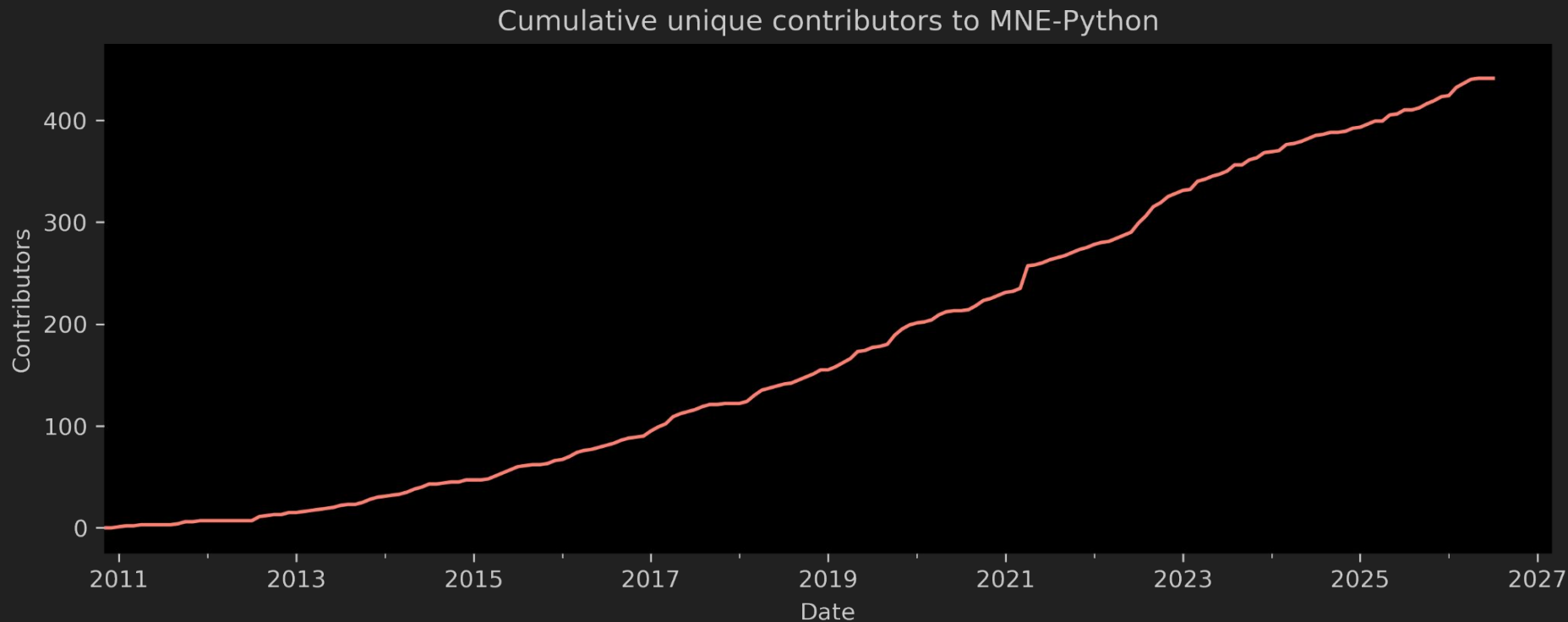
Up next: Curriculum

- Maintainer onboarding lessons
 - Eventually integrated into learn.scientific-python.org
- Contributor ladder

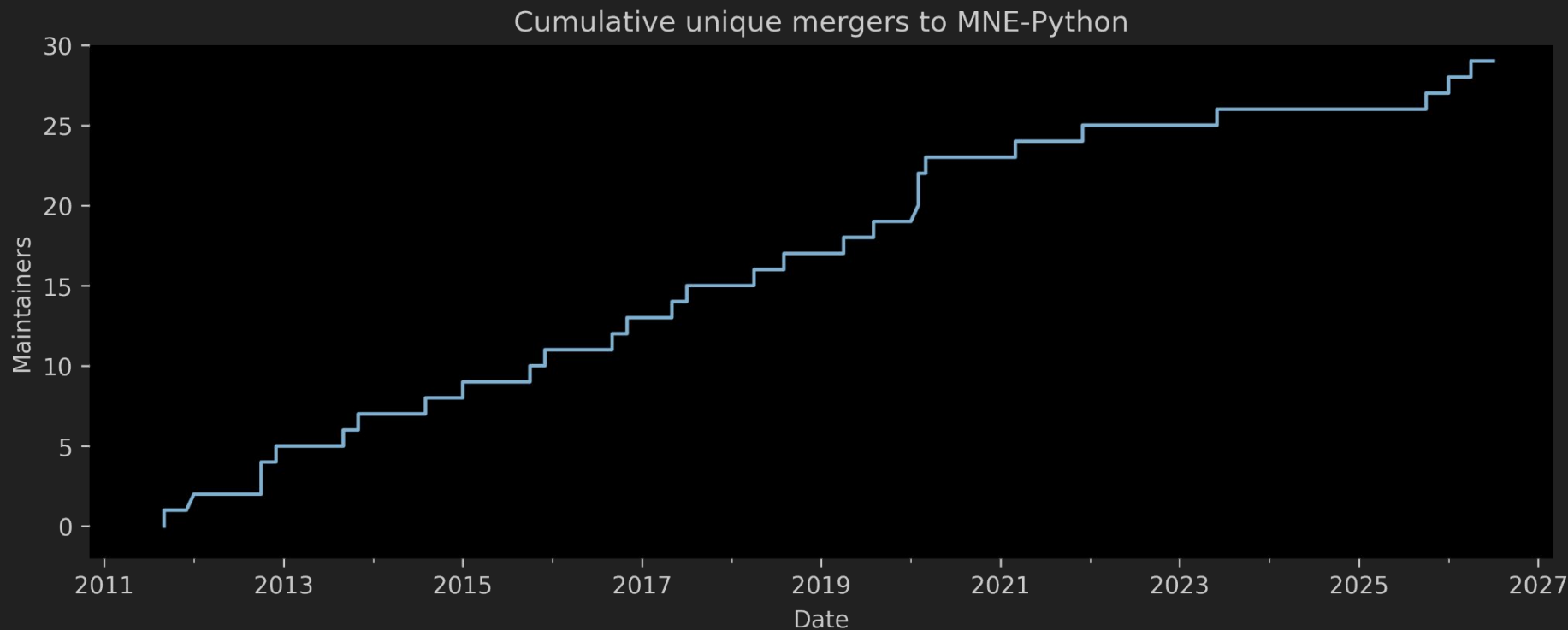
Why is MNE-Python struggling?

It's not just a pipeline problem

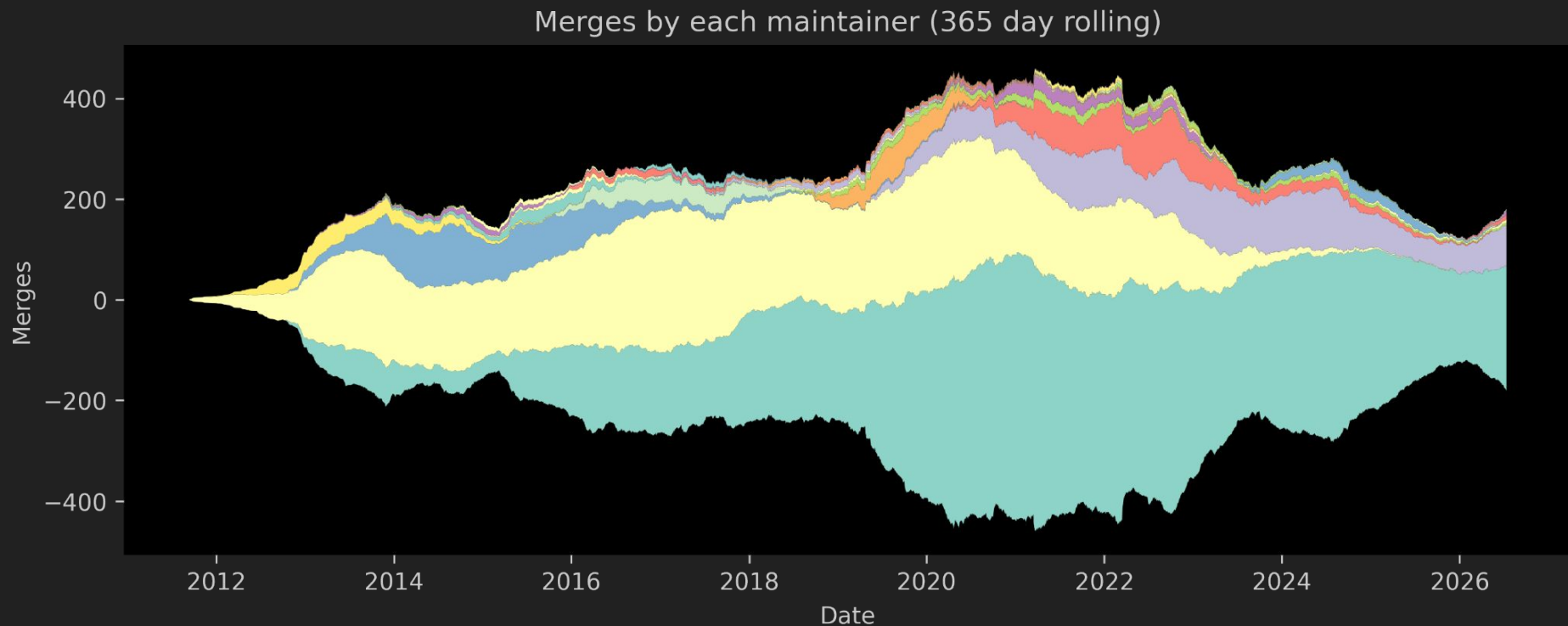
Contributor growth has been steady...



Contributors are becoming maintainers...



...but maintainers are drifting away



Why?

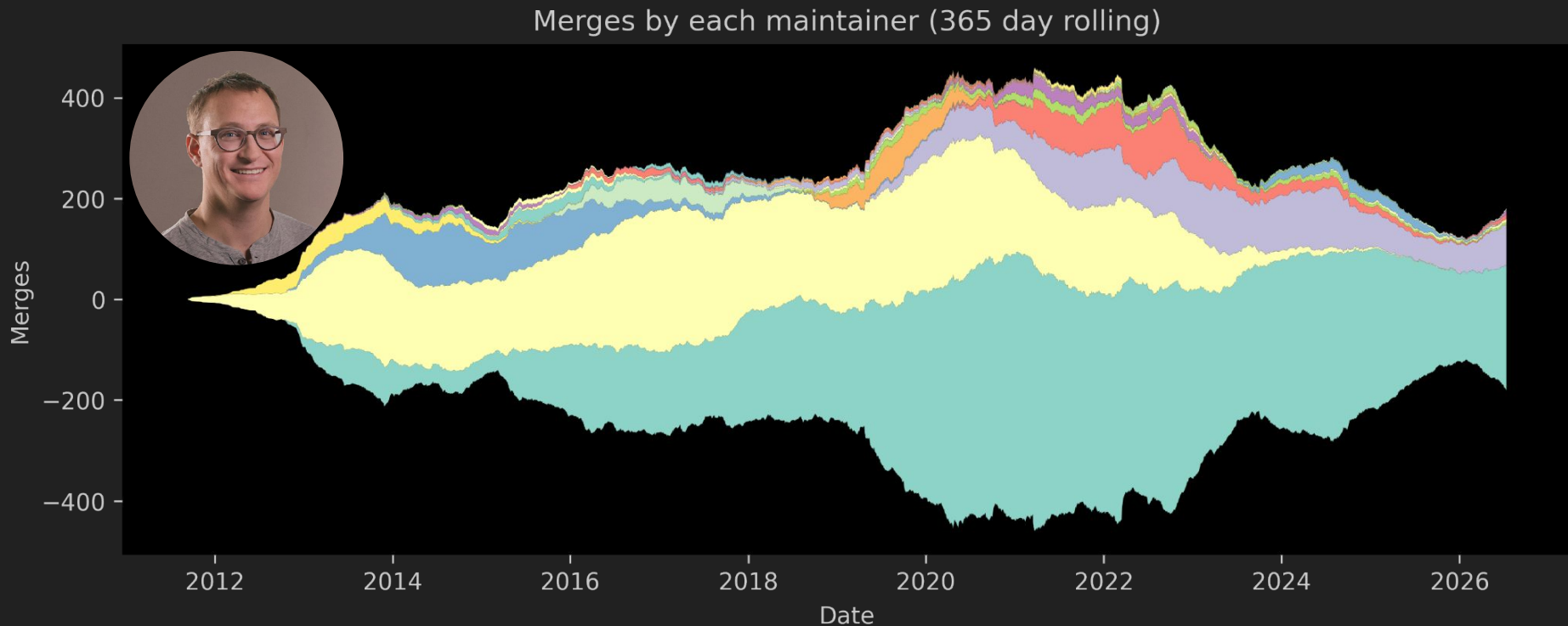
Why?

Career transitions (academia → industry)?

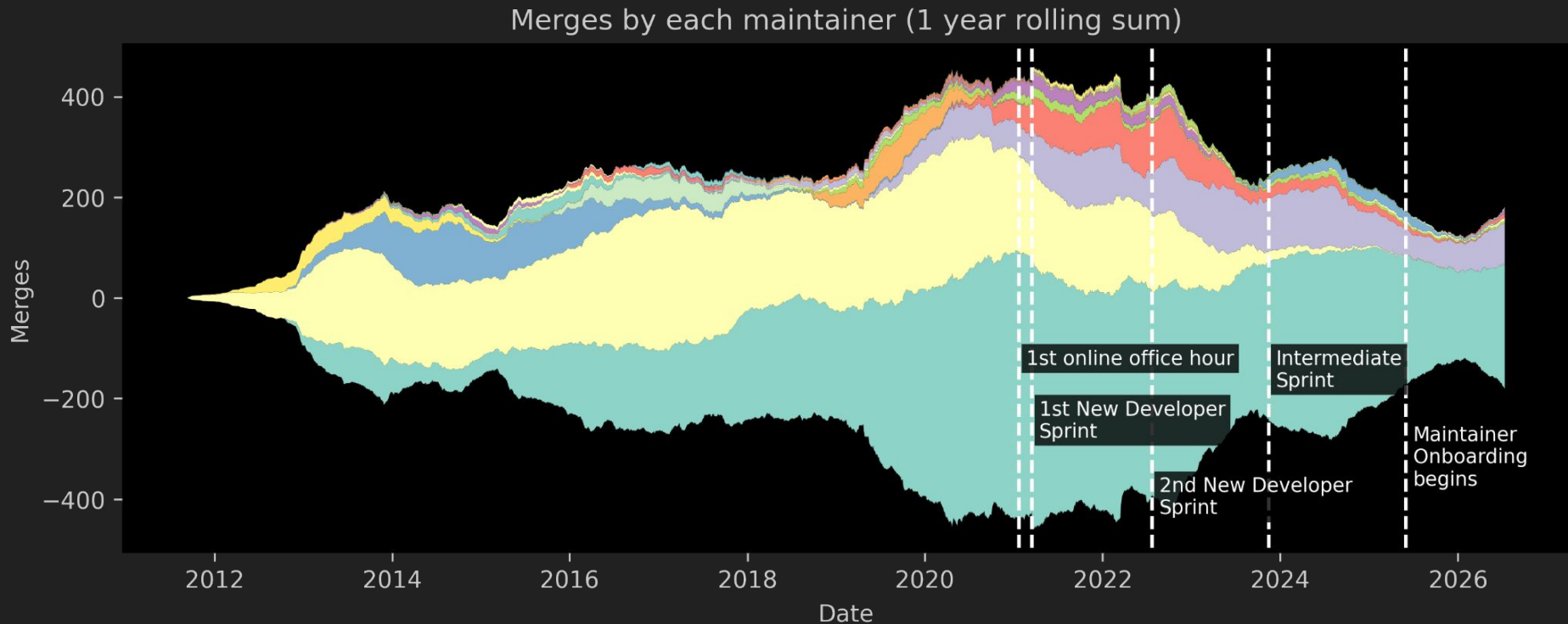
Unhappy with new governance?

Busy with new, post-COVID hobbies?

We relied too much on project founder's lab



Now: trying to do it bottom-up



Summary and outlook

- I think it's working (time will tell)
- It was tremendously expensive and slow
- Hopefully the curriculum will help
- Maybe AI will ultimately relieve some of the pressure?

Acknowledgments



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Evolution of our governance

